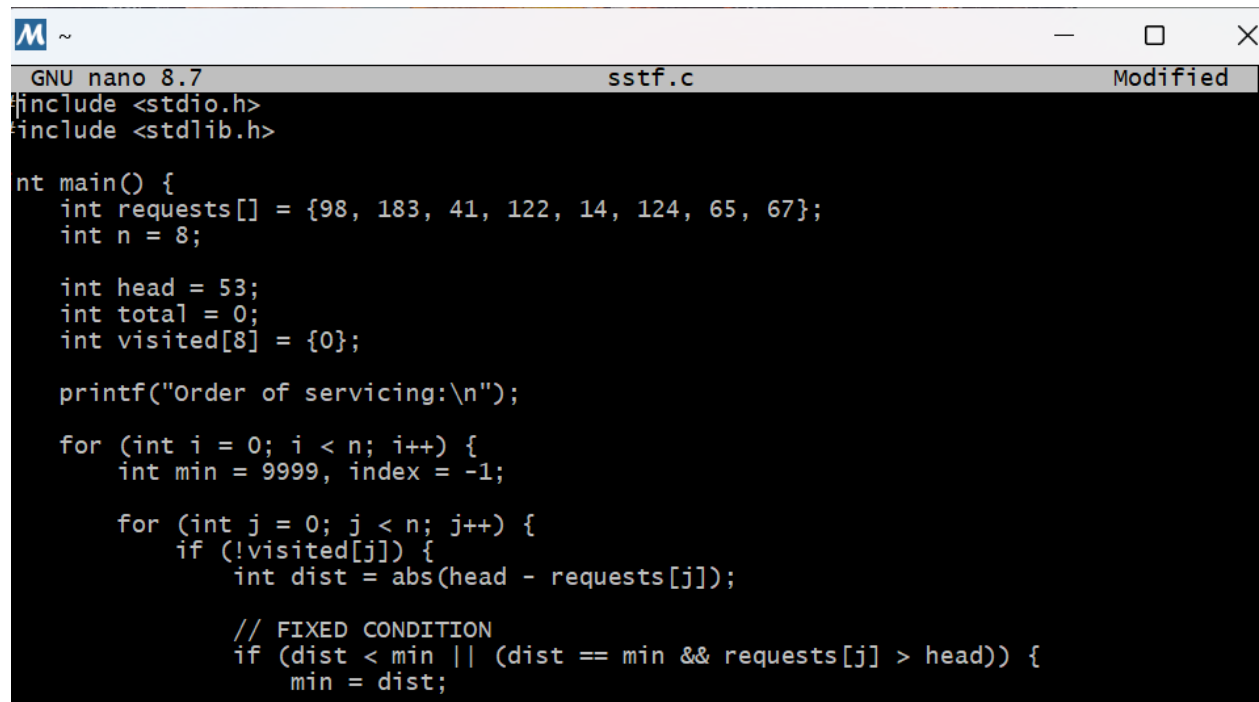


10.[Disk Scheduling]

Consider a disk queue that contains requests for input/output operations at cylinder numbers **98, 183, 41, 122, 14, 124, 65, and 67**. The disk head is initially positioned at **cylinder 53**. The cylinders are numbered from **0 to 199**. Use the **Shortest Seek Time First (SSTF)** disk scheduling algorithm to service these requests. Write a program to determine the **total head movement**, measured in the number of cylinders traversed, while processing all the given requests.

CODE :



```
GNU nano 8.7 sstf.c Modified
#include <stdio.h>
#include <stdlib.h>

int main() {
    int requests[] = {98, 183, 41, 122, 14, 124, 65, 67};
    int n = 8;

    int head = 53;
    int total = 0;
    int visited[8] = {0};

    printf("Order of servicing:\n");

    for (int i = 0; i < n; i++) {
        int min = 9999, index = -1;

        for (int j = 0; j < n; j++) {
            if (!visited[j]) {
                int dist = abs(head - requests[j]);

                // FIXED CONDITION
                if (dist < min || (dist == min && requests[j] > head)) {
                    min = dist;
                }
            }
        }

        printf("%d ", requests[index]);
        visited[index] = 1;
        total += min;
        head = requests[index];
    }

    printf("\nTotal head movement: %d", total);
}
```

```
GNU nano 8.7 sstf.c Modified
    if (!visited[j]) {
        int dist = abs(head - requests[j]);

        // FIXED CONDITION
        if (dist < min || (dist == min && requests[j] > head)) {
            min = dist;
            index = j;
        }
    }

    total += min;
    head = requests[index];
    visited[index] = 1;

    printf("%d -> ", head);
}

printf("\nTotal Head Movement = %d\n", total);

return 0;
}
```

OUTPUT :

```
ASUS@Kalash-Laptop MINGW64 ~
$ nano sstf.c

ASUS@Kalash-Laptop MINGW64 ~
$ gcc sstf.c -o sstf

ASUS@Kalash-Laptop MINGW64 ~
$ ./sstf
Order of servicing:
65 -> 67 -> 41 -> 14 -> 98 -> 122 -> 124 -> 183 ->
Total Head Movement = 236

ASUS@Kalash-Laptop MINGW64 ~
$
```